

A database on flexural and shear strength of reinforced recycled aggregate concrete beams and comparison to Eurocode 2 predictions



Nikola Tošić*, Snežana Marinković, Ivan Ignjatović

University of Belgrade, Faculty of Civil Engineering, Bulevar kralja Aleksandra 73, 11000 Belgrade, Serbia

HIGHLIGHTS

- Database of experimental results on recycled aggregate concrete beams compiled.
- Database filtered by different parameters—concrete strength, anchorage, etc.
- Database analyzed to identify clear failure types—flexural or shear.
- Sub-databases formed for flexural and shear failure (with and without stirrups).
- Applicability of Eurocode 2 provisions to recycled aggregate concrete beams tested.

ARTICLE INFO

Article history:

Received 20 June 2016

Received in revised form 29 September 2016

Accepted 6 October 2016

Keywords:

Recycled aggregate concrete

Reinforced concrete

Beams

Database

Flexural strength

Shear strength

Eurocode 2

ABSTRACT

A comprehensive database of recycled aggregate concrete and companion natural aggregate concrete beams' flexural and shear strength was compiled from 217 experimental results. Strict criteria were applied to determine the failure type. Sub-databases were formed with beams failing in flexure and shear with and without stirrups. On each sub-database the applicability of Eurocode 2 provisions for flexural and shear strength to recycled aggregate concrete beams was tested. The results show that flexural and shear strength of recycled aggregate concrete beams without stirrups is successfully predicted by Eurocode 2. As for beams with stirrups, further research and experimental results are necessary.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

The construction industry today faces urgent calls to reform. The current rate of consumption of natural resources, waste generation and greenhouse gas emissions is unsustainable. On the one hand, new concrete requires the use of natural river or crushed stone aggregates, up to 15 billion tons annually worldwide [1]. On the other hand, old concrete structures are demolished and construction and demolition (C&D) waste is generated in large quantities, around 850 million tons in the EU annually [2].

It is not surprising that alternatives are being sought out. One solution that solves both problems simultaneously is recycling of concrete waste. Through a process that usually involves multi-

stage crushing, eliminating impurities and sieving, a new aggregate is produced called recycled concrete aggregate (RCA). When this new aggregate is used to make concrete, with complete or partial replacement of natural aggregate, this concrete is called recycled aggregate concrete (RAC).

Recycled concrete aggregate and recycled aggregate concrete have been studied for several decades [3]. At the material level, practically all important characteristics of RCA and RAC have been studied, from short-term and long-term mechanical properties to durability [4–8]. The main characteristic that distinguishes RCA from natural aggregate is the certain quantity of cement paste that remains attached to the aggregates after crushing. This residual cement paste is the reason for higher water absorption of RCA compared with natural aggregates, especially in the case of fine RCA [9,10]. Beside the empirical observations about the influence of higher RCA water absorption on RAC properties, there have also been deeper, fundamental studies that demonstrated how the

* Corresponding author.

E-mail addresses: ntosic@imk.grf.bg.ac.rs (N. Tošić), sneska@imk.grf.bg.ac.rs (S. Marinković), ivani@imk.grf.bg.ac.rs (I. Ignjatović).

moisture state and water absorption of RCA influence the evolution of cement hydration [11]. The high water absorption of fine RCA has led to them mostly being avoided when producing RAC. However, even for coarse RCA the situation isn't much better as they make up only 1% of aggregates being used in structural concrete production worldwide [12].

This doesn't mean that research into the structural application of RAC has been lacking. Beside investigations of short-term flexural and shear performance of reinforced RAC beams, which are studied in this paper, there has been significant research on various other topics such as semi-precast RAC elements [13], shaking-table and pushover analyses of complete RAC frame structures [14,15] and long-term behavior of RAC beams [16]. Important literature also exists on the ecological and economic viability of RCA production and use [17–19].

Despite all of this, coordinated efforts by national and international institutions and organizations to codify the design procedures for RAC structural members have been lacking. Code provisions for material properties of RAC have been successfully tested and proven to be applicable [20,21] but these results cannot simply be extrapolated onto structural members. With the exception of China and its Technical Code on the Application of Recycled Concrete [22], neither European nor American concrete or standardization institutes have integrated provisions for the design of RAC structural members into their respective codes [23,24], even though researchers have attempted to demonstrate design procedures of RAC members according to them [25]. Beside natural aggregate concrete (NAC), only high-strength and lightweight aggregate concretes have been dealt with in their codes. Consequently, practicing engineers are faced with uncertainties in the rare situations when they have the opportunity to design structural RAC members.

In the present paper, results on short-term flexural and shear behavior of RAC beams were gathered from available literature. Strict selection criteria were applied to determine the failure type, flexure or shear. A comprehensive database was compiled with three sub-databases: beams failing in flexure, in shear without and with stirrups. These selected results can be considered to represent well-executed experiments and clear failure types with as little shear-flexure interaction as possible. The compilation of such a database has been missing from existing literature and is critical for any design formula verification and calibration.

As a second part of this study, EN 1992-1-1:2004 (Eurocode 2 or EC2) [23] provisions for predicting flexural and shear strength were tested on RAC beams by calculating the ratio of test-to-predicted flexural and shear strengths. This ratio was called the “model factor” γ , as it represents the uncertainty and variability introduced into calculations by the model itself and by its appropriateness. This is separate from the uncertainties arising from loads and material properties, covered in design by the partial safety factors which were removed and characteristic values of material properties were replaced with mean values. This approach is, in essence, the same as that proposed by EN 1990:2002 (Eurocode – Basis of structural design) in Annex D—Design assisted by testing [26].

The accuracy and precision of EC2 provisions was assessed using qualitative and quantitative analyses. In this study, accuracy is understood as the closeness of the model factor's mean value to 1.0 and precision is determined by the value of the model factor's coefficient of variation (CoV), i.e. scatter.

2. Database formation

2.1. Selection of studies

The first step in this research was the collection of all available studies on shear and flexural strength of RAC beams. A review of

existing literature yielded 16 studies [27–42] carried out in the period from 2001 to 2015 with a total of 217 experimental results. All of the studies were comparative tests of RAC and NAC beams. The replacement ratios of natural aggregate by coarse RCA, chosen for this study, were 0, 50, and 100%, i.e. NAC, RAC50, and RAC100 concretes. In studies [32,33,35] the replacement ratio of 63.5% was assigned to RAC50 and the replacement ratio of 74.3% was assigned to RAC100 concrete.

Before compiling any database, rigorous selection criteria had to be established by which results would be tested. Since the aim of the study was to test the applicability of EC2 [23] flexural and shear strength predictions on RAC beams, the selection criteria had to ensure that only well-executed experiments and unambiguous results entered the database.

Only slender beams were analyzed since the test results on non-slender RAC beams are scarce. An initial screening was performed and any beams with a shear span-to-effective depth ratio smaller than 2.4 were eliminated. This value was chosen as critical so that a comparison with other databases could be performed [43,44]. This eliminated 17 results. Since EC2 prescribes different formulas for concrete classes greater than C50/60 and since high-strength RAC is not very common, only concretes with strengths smaller than 63 MPa were considered. This eliminated another 3 results. If the beams had stirrups then the minimum transverse reinforcement ratio was checked according to the EC2 limit:

$$\rho_w \geq 0.08 \sqrt{f_c / f_{yw}} \quad (1)$$

where

ρ_w – transverse reinforcement ratio

f_c – 28-day concrete compressive strength on a Ø150/300 mm cylinder (MPa)

f_{yw} – transverse reinforcement yield strength (MPa)

This criterion eliminated another 3 results. Finally, 194 experimental results on NAC, RAC50, and RAC100 beams were left. Data were collected on beam geometry (width, depth, and effective depth), shear span-to-effective depth ratio, longitudinal and transverse reinforcement ratios and yield strength, concrete properties (percentage of RCA, maximum aggregate size, and compressive strength), and beam shear and flexural strengths. The data were then entered into an Excel spreadsheet that can be found in [Appendix A](#).

2.2. Anchorage and shear-flexure interaction checks

Although practically all of the studies claim to be testing either flexural or shear strength of beams, this cannot be trusted at face value. It is not uncommon for researches investigating shear strength to report a flexural failure of beams or vice versa. This means that the experimental setup and failure load for each beam have to be checked for anchorage failure and shear-flexure interaction.

To check against anchorage failure, the following condition must be satisfied:

$$\beta_{lb} = l_{b,req} / l_{b,prov} \leq 1 \quad (2)$$

where $l_{b,req}$ and $l_{b,prov}$ are the required and provided anchorage lengths (in mm) and β_{lb} is the anchorage criterion. The required anchorage length was calculated according to section 8.4 of EC2 as:

$$l_{b,req} = \alpha_1 \alpha_2 \alpha_3 \alpha_4 \alpha_5 l_{b,reqd} = 0.7 \cdot \frac{\sigma_s}{4} \cdot \frac{\sigma_s}{2.25 \eta_1 \eta_2 f_{ct}} = 0.7 \cdot \frac{\sigma_s}{9} \cdot \frac{\sigma_s}{f_{ct}} \quad (3)$$

where

α_1 – α_5 – coefficients taking into account the shape of the bars, concrete cover, confinement by transverse reinforcement

(welded and not welded to longitudinal reinforcement), and confinement by transverse pressure

$\emptyset$ – maximum diameter of longitudinal reinforcement (mm)

σ_s – stress in longitudinal reinforcement at start of anchorage length (MPa)

η_1 – η_2 – coefficients taking into account the bond condition and reinforcement diameter

f_{ct} – 28-day concrete axial tensile strength (MPa)

While in some studies the bars had hooks and in others they were straight, all of the studies used steel support plates and consequently introduced large transverse pressures at the supports. Because of this the product $\alpha_1\alpha_2\alpha_3\alpha_4\alpha_5$ was taken as the minimum allowed value of 0.7 in all cases. All the studies had good bond conditions and bars with diameters smaller than 32 mm so the product $\eta_1\eta_2$ was equal to 1. The concrete tensile strength was calculated from compressive strength according to the formula given in Table 3.1 of EC2:

$$f_{ct} = 0.3 \cdot (f_c)^{2/3} \quad (4)$$

As for the stress in the longitudinal reinforcement, the calculation depended on whether the beam had stirrups or not since the mechanical models are different. In the case of beams with stirrups, the usual truss model was adopted and the stress calculated according to clause 6.2.3(7) of EC2:

$$\sigma_s = \frac{0.5 \cdot V_{R,test} \cdot \cot \theta}{A_{sl}} = \frac{1.25 \cdot V_{R,test}}{A_{sl}} \quad (5)$$

where

$V_{R,test}$ – experimental value of shear strength (N)

A_{sl} – longitudinal reinforcement area (mm²)

θ – angle of concrete compression strut inclination

When calculating Eq. (5) the angle θ was conservatively taken as the minimum value of 21.8° according to EC2. Mechanically, θ represents the angle of the concrete compression strut inclination in the truss model; in principle, it depends on the amount of stirrups. Hence, adopting $\theta = 21.8^\circ$ in Eq. (8) is a conservative and simplistic assumption.

In the case of beams without stirrups, the load transfer mechanism is different so another model was necessary. For this purpose the provision given in Model Code 2010 (MC2010), equation (7.3–18) was adopted [45]:

$$\sigma_s = \frac{V_{R,test}}{A_{sl}} \quad (6)$$

It should be noted that in Eqs. (5) and (6) a simpler and more conservative assumption would have been to assume yielding of the longitudinal reinforcement, i.e. $\sigma_s = f_{yl}$. Nonetheless, as can be seen from Appendix A, all 194 results satisfy the anchorage criterion.

In the case of shear-flexure interaction, however, the situation is a little more complicated. Since the aim of the study was to analyze EC2 predictions of shear and flexural strength on RAC beams, the database had to be filtered for results that exemplified true and clear shear or flexural failures. A similar approach was taken in [43,44] where a check for flexural failures was performed on beams that were stated to have failed in shear. The check performed in [43,44] consisted of calculating the test-to-predicted flexural strength ratio and checking if it is smaller than 1.1. If so, the beam was deemed to have failed in shear since it did not surpass its flexural strength by more than 10%.

However, using this approach, some flexural failures can be classified as shear and some situations in which the failure type is unclear, can be classified as either one. Consider, for example,

a beam with a test-to-predicted flexural strength ratio equal to 1.05 and a test-to-predicted shear strength ratio equal to 0.65. Using this criterion, the beam would be classified as failing in shear, though it most likely failed in flexure. Another problematic situation would be a beam with both test-to-predicted strength ratios equal to 1.05. Again, it would be classified as a shear failure, even though it is actually very difficult to determine a clear failure type in this situation.

In order to overcome this problem, a slightly different approach was formulated in the current study. First, the test-to-predicted flexural and shear strength ratios, β_{fl} and β_{sh} respectively, were calculated. When choosing according to which model to calculate the predicted values, care had to be taken to select the most accurate and physically meaningful models.

For flexural strength, the standard procedure given in MC2010 was thought to be satisfactory. For the concrete stress-strain relation the parabola-rectangle diagram was chosen whereas for the reinforcement steel stress-strain relation the idealized bi-linear diagram with a horizontal top branch was selected. The predicted flexural strength was calculated as:

$$M_{R,pred} = A_{sl} f_{yl} d \left(1 - 0.513 \frac{A_{sl} f_{yl}}{b d f_c} \right) \quad (7)$$

where

$M_{R,pred}$ – predicted value of flexural strength (Nm)

f_{yl} – longitudinal reinforcement yield stress (MPa)

d – cross-section effective depth (mm)

b – cross-section width (mm)

For shear strength, MC2010 was chosen, specifically the level III approximation [45]. It was chosen as the physically most meaningful and justifiable model, based on the Modified compression field theory (MCFT). MC2010 defines shear strength as:

$$V_{R,pred} = V_{R,c} \quad (8)$$

for beams without stirrups and for beams with stirrups greater than the minimum defined by Eq. (1) as:

$$V_{R,pred} = \begin{cases} V_{R,c} + V_{R,s} & \text{for } V_{R,c} + V_{R,s} < V_{R,max} \\ \max(V_{R,c}; V_{R,s}) \leq V_{R,max} & \text{for } V_{R,c} + V_{R,s} \geq V_{R,max} \end{cases} \quad (9)$$

where

$V_{R,pred}$ – predicted value of shear strength (N)

$V_{R,c}$ – shear strength attributed to concrete (N)

$V_{R,s}$ – shear strength provided by stirrups (N)

$V_{R,max}$ – maximum allowed shear strength (N)

The shear strengths defined in Eq. (8) and (9) were calculated according to the following expressions:

$$V_{R,c} = k_v \sqrt{f_c} z b_w \quad (10)$$

$$V_{R,s} = \frac{A_{sw}}{s} z f_{yw} \cot \theta \quad (11)$$

$$V_{R,max} = k_e \eta_{fc} f_c z b_w \sin \theta \cos \theta \quad (12)$$

where

z – inner lever arm = 0.9· d (mm)

b_w – cross-section width or web width for I, L and T sections (mm)

A_{sw} – transverse reinforcement area (mm²)

s – transverse reinforcement spacing (mm)

The remaining coefficients and parameters were determined from the following equations:

$$\eta_{fc} = \left(\frac{30}{f_c}\right)^{1/3} \leq 1.0 \quad (13)$$

$$k_e = \frac{1}{1.2 + 55\varepsilon_1} \leq 0.65 \quad (14)$$

$$\varepsilon_1 = \varepsilon_x + (\varepsilon_x + 0.002)\cot^2\theta \quad (15)$$

$$\varepsilon_x = \frac{V_{E,test} \left[\frac{d}{z} \left(\frac{a}{d} - 1 \right) + 1 \right]}{2E_s A_{sl}} \quad (16)$$

$$\theta = 20^\circ + 10000\varepsilon_x \quad (17)$$

$$k_v = \begin{cases} = \frac{0.4}{1+1500\varepsilon_x} \cdot \frac{1300}{1000+k_{dg}z} & \text{if } \rho_w = 0 \\ = \frac{0.4}{1+1500\varepsilon_x} \cdot \left(1 - \frac{V_{E,test}}{V_{R,max,pred}}\right) \geq 0 & \text{if } \rho_w \geq 0.08\sqrt{f_c}/f_{yw} \end{cases} \quad (18)$$

$$k_{dg} = \frac{32}{16 + d_g} \geq 0.75 \quad (19)$$

where

ε_x – longitudinal strain at mid-depth of beam (mm/mm)
 E_s – reinforcement steel modulus of elasticity (N/mm²)
 d_g – maximum aggregate size (mm)

The test-to-predicted flexural and shear strength ratios were then calculated as:

$$\beta_{fl} = M_{R,test}/M_{R,pred} \quad (20)$$

$$\beta_{sh} = V_{R,test}/V_{R,pred} \quad (21)$$

as seen in columns 20 and 25 in Appendix A. The next step was to determine how the failure type can be identified with as much certainty as possible. Nominally, a test-to-predicted strength ratio greater than 1 points to a failure type. However, situations where both ratios are greater than 1 or smaller than 1 are also possible. It is clear that what points to a failure type isn't the absolute value of a test-to-predicted strength ratio but rather the difference between the two. The only outstanding question is then the selection of the critical value of this difference in reference to which failure types would be identified. The ratios should be sufficiently apart to guarantee that there is as little shear-flexure interaction as possible.

One approach to this problem would be defining a joint probability distribution of the difference $\Delta = \beta_{sh} - \beta_{fl}$ and operating with it. These calculations can be further complicated depending on the correlation between the variables and their marginal probability distributions. Instead, in this study, an empirical approach was chosen. First, a critical value, Δ_{cr} , was chosen on the basis of experience. Secondly, the complete analysis was carried out using this criterion. Finally, the robustness and validity of the analysis and conclusions were tested by carrying out a sensitivity analysis of the critical value Δ_{cr} .

From previous studies [38,46], it was found that the CoVs of β_{fl} and β_{sh} , calculated according to different codes, are in the range of 0.05–0.15 and 0.20–0.30 respectively. Using these values as a reference point, the critical value Δ_{cr} was chosen as 0.35 ($= \text{CoV}_{\text{shear}} + \text{CoV}_{\text{flexure}} = 0.25 + 0.10$).

After calculating Δ_{cr} for each beam in the database the results were sorted into three sub-databases. If $\Delta_{cr} \geq 0.35$ and the beam had stirrups, the result was assigned to database Shear S and if it had no stirrups it was assigned to database Shear NS. If $\Delta_{cr} \leq -0.35$ the results were assigned to database Flexure. If $-0.35 \leq \Delta_{cr} \leq 0.35$ the result was left out of all databases. In this way, out of the original 194 results, 49 were assigned to database Flexure, 69 to Shear NS,

and 25 to Shear S. This means that 51 beams were excluded from all databases since according to the selected criteria it was not possible to determine whether the failure was shear or flexural.

For a more detailed presentation of the experimental results, Figs. 1–4 are given. In each of the figures, the number of beams n is plotted versus a certain parameter—concrete compressive strength f_c , longitudinal reinforcement ratio ρ_l , beam effective depth d , and shear span-to-effective depth ratio a/d , respectively. Each parameter is divided into classes and the number of beams in each class is plotted, given separately for beams assigned to a database (Flexure, Shear NS, or Shear S) and for beams unassigned to any sub-database.

From Fig. 1 it can be seen that concrete compressive strengths in the range of 30–45 MPa comprise 72% of the original database. Also interestingly, most of the results on higher compressive strengths (>55 MPa) remained unassigned to any sub-database.

Fig. 2 shows that longitudinal reinforcement ratios 0.5–2% make up 64% of the original database. However, there is a spike in the number of beams with a 2.5–3% longitudinal ratio (15% of the results) and all of them were assigned to a sub-database.

In Fig. 3, the number of beams is plotted versus the beams' effective depth and 92% of the results are with $d < 400$ mm. Importantly, the highest number of unassigned results is for beams with $d < 250$ mm, while all of the beams with $d > 400$ mm were assigned to sub-databases.

Finally, from Fig. 4 an almost uniform distribution of shear span-to-effective depth ratios between 2.4 and 4.4 can be seen. The largest number of unassigned results is in the 2.4–3.2 range.

In order to further expand the number of results, other databases available in literature were analyzed. The ACI-DAFStb database of shear strength of NAC beams contains 744 results of shear strength of slender beams without stirrups and 87 results on beams with stirrups [43,44]. Applying the criteria described in this section 507 results were assigned to database Shear NS and 37 to database Shear S.

In total, this amounts to 49 results in database Flexure (18 NAC, 14 RAC50, and 17 RAC100 beams), 576 results in Shear NS database (530 NAC, 24 RAC50, and 22 RAC100), and 62 results in Shear S (45 NAC, 8 RAC50, and 9 RAC100).

3. Eurocode 2 flexural and shear strength predictions for RAC beams

3.1. Flexural strength

In this section the predictive capability of EC2 provisions for flexural strength of RAC beams was tested. For all the results in

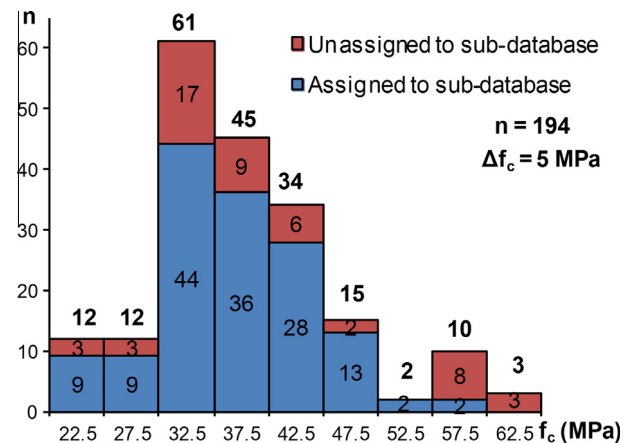


Fig. 1. Number of beams n plotted versus concrete compressive strength f_c .

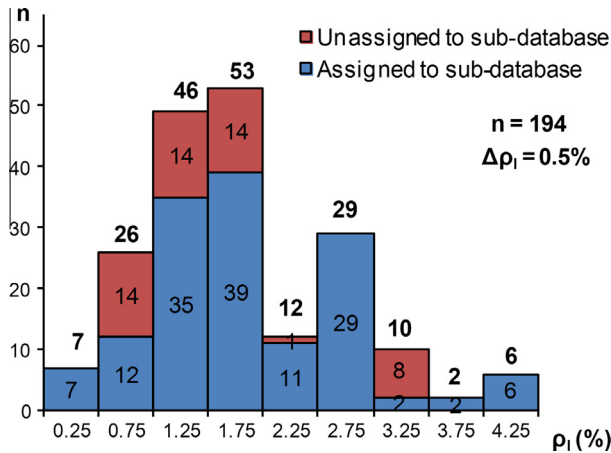


Fig. 2. Number of beams n plotted versus the longitudinal reinforcement ratio ρ_l .

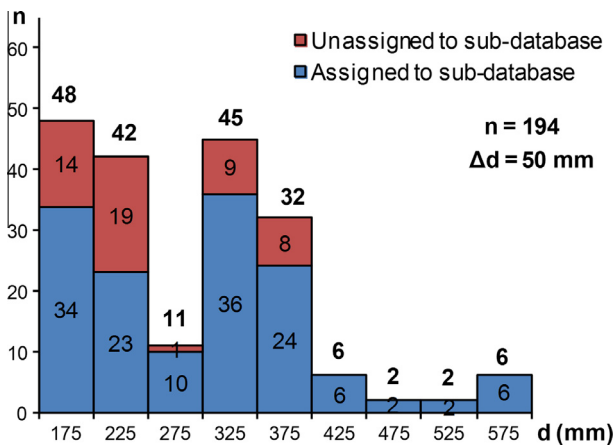


Fig. 3. Number of beams n plotted versus beam effective depth d .

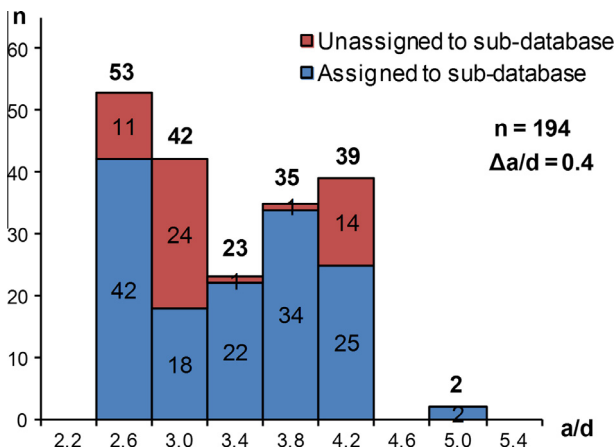


Fig. 4. Number of beams n plotted versus the shear span-to-effective depth ratio a/d .

database Flexure the EC2 predictions were calculated. For concrete, a parabola-rectangle stress-strain diagram was chosen and for reinforcement steel, a bi-linear stress-strain diagram with a horizontal top branch. Since the EC2 provisions for flexural strength are identical to those of MC2010, the predicted flexural strength was calculated according to Eq. (7) given in the previous section.

The database Flexure along with relevant data and the model factor γ_{fl} is given in Table 1 and the statistical descriptors are given

in Table 2. The mean values for all three samples (NAC, RAC50, and RAC100) are very close to 1, below 1.1, and the CoVs are satisfactorily low as well. This is to be expected as the analytical model for flexural failure is well-established and physically meaningful.

The next step was to visually assess the results, plotting the model factor values against relevant parameters, Figs. 5–7. In all of the figures horizontal lines were plotted representing the 5–95% interval around the mean value for NAC beams ($\mu \pm 1.645 \cdot \sigma$) for easier assessment of the fit between RAC and NAC beams. As expected for flexure, this 5–95% interval is narrow and practically all the results fit within it. Moreover, no correlation emerges between the model factor and any of the parameters—concrete compressive strength, longitudinal reinforcement ratio or cross-section effective depth. This means that the model's predictive capability is equal in the complete range of the parameters' values.

The initial visual inspection pointed to an excellent agreement between RAC and NAC beams, so further calculations were performed to quantify this observation. The statistical descriptors given in Table 2 can be used for statistical tests and comparisons of RAC and NAC beams. The usual procedure in these cases is to carry out the so-called t -test and compare the means of different samples.

When dealing with relatively small sample sizes, as in this case, the t -test requires the tested samples to be normally distributed [47]. To determine this, the Kolmogorov-Smirnov goodness-of-fit test was carried out. This is a non-parametric test that quantifies the distance between an empirical distribution function of the sample and the cumulative distributive function of the Normal distribution. In the case of NAC beams the following hypotheses were tested:

Null hypothesis H_0 : The distribution of $\gamma_{fl,NAC}$ is Normal with $\mu = 1.064$ and $\sigma = 0.092$

Alternate hypothesis H_1 : $\gamma_{fl,NAC}$ has a different distribution

Level of significance: $\alpha = 0.05$

RAC50 and RAC100 beams were tested in the same way for their descriptors as given in Table 2. The test statistic of the Kolmogorov-Smirnov test is the maximum difference between the empirical and hypothesized distributions and it is compared to a critical value depending on the significance level and sample size. If the test statistic is smaller than the critical value, the null hypothesis should be retained. The test statistics were 0.118 for NAC, 0.261 for RAC50, and 0.245 for RAC100 beams and the critical values were 0.309, 0.349, and 0.318, respectively. This means that at the significance level $\alpha = 0.05$ (the probability that a test will reject a null hypothesis that is actually true) the null hypothesis should be retained for all three concretes.

The final step was to perform the t -test and see whether the means of γ_{fl} for RAC50 and RAC100 were significantly different from the mean of γ_{fl} for NAC beams. Both for RAC50 and RAC100 the following hypotheses were tested:

Null hypothesis H_0 : $\mu_{NAC} = \mu_{RAC50/100}$

Alternate hypothesis H_1 : $\mu_{NAC} \neq \mu_{RAC50/100}$

Level of significance: $\alpha = 0.05$

The t -test uses the sample means and variances (σ^2) to calculate a test statistic t that follows the Student's T distribution (hence the name, t -test). The test statistic and the cumulative distribution function are used to calculate the so-called p -value which, if smaller than 0.05 (the significance level), points to a significant difference between the samples, i.e. the null hypothesis should be rejected.

The calculated p -values were 0.734 for the NAC-RAC50 comparison and 0.524 for the NAC-RAC100 comparison. This means that there is no significant difference between the NAC and RAC

Table 1

Eurocode 2 predictions of flexural strength for 49 selected beams.

Study	Specimen	RCA (%)	b_w (mm)	d (mm)	ρ_l (%)	f_{yt} (MPa)	f_c (MPa)	$M_{E, test}$ (kNm)	$M_{R, pred}$ (kNm)	γ_{fl}
[29]	V45-03-WB	0	150	160	1.06	331	57.0	15.0	13.6	1.10
	VEX45-03-WB		150	160	1.06	331	55.3	15.3	13.6	1.13
	V-01-10WB		150	160	0.59	331	30.6	8.0	8.0	1.00
	V-01-10DB		150	160	0.59	331	32.5	9.1	8.0	1.13
[32]	CL-Av	0	200	304	1.99	420	37.1	142.7	165.1	0.86
	CG-Av		200	304	1.99	420	33.8	139.1	162.4	0.86
[36]	BSF4-A0		400	525	2.34	380	26.9	878.9	813.8	1.08
[37]	NAC1a		200	268	0.28	640	35.0	28.4	25.2	1.13
	NAC2a	0	200	263	1.46	550	35.0	108.6	97.5	1.11
	NAC3a		200	244	2.54	550	35.0	137.6	134.8	1.02
[40]	F0-1a		150	200	1.3	572	38.6	42.6	41.3	1.03
	F0-1b		150	200	1.3	572	38.6	43.1	41.3	1.04
	F0-2a		150	200	1.3	572	46.5	43.8	42.1	1.04
	F0-2b		150	200	1.3	572	46.5	43.8	42.1	1.04
[41]	N0-0.5	0	135	230	0.5	377	38.6	15.9	13.3	1.20
	N0-1.0		135	230	1	408	38.6	28.2	23.8	1.19
	N0-1.5		135	230	1.5	389	38.6	36.9	33.6	1.10
	N0-1.8		135	230	1.8	410	38.6	52.8	48.1	1.10
[37]	RAC50-1a	50	200	268	0.28	640	35.4	27.0	25.2	1.07
	RAC50-2a		200	263	1.46	550	35.4	110.6	97.6	1.13
	RAC50-3a		200	244	2.54	550	35.4	160.4	135.2	1.19
[40]	F50-1a	50	150	200	1.3	572	40.0	41.8	41.5	1.01
	F50-1b		150	200	1.3	572	40.0	43.1	41.5	1.04
	F50-2a		150	200	1.3	572	39.3	41.3	41.4	1.00
	F50-2b		150	200	1.3	572	39.3	41.3	41.4	1.00
[41]	N50-0.5	50	135	230	0.5	377	29.0	13.6	13.2	1.03
	N50-1.0		135	230	1	408	29.0	24.4	23.4	1.04
	N50-1.5		135	230	1.5	389	29.0	32.8	32.8	1.00
	N50-1.8		135	230	1.8	410	29.0	50.5	46.4	1.09
[32]	EM-Min	63.5	200	304	0.49	420	41.6	46.0	29.5	1.56
	EM-Av		200	304	1.99	420	41.6	149.2	168.1	0.89
	EM-Max		200	304	3.26	420	41.6	221.9	208.7	1.06
[32]	EV-Min	74.3	200	304	0.49	420	49.1	46.7	29.6	1.58
	EV-Av		200	304	1.99	420	49.1	150.2	171.9	0.87
	EV-Max		200	304	3.26	420	49.1	225.2	215.1	1.05
[29]	CR45-03-WB	100	150	160	1.06	331	46.5	14.8	13.5	1.10
	CREX45-03-WB		150	160	1.06	331	46.6	15.1	13.5	1.12
	CR45-01-10WB		150	160	0.59	331	30.4	8.5	8.0	1.06
	CR45-01-10DB		150	160	0.59	331	28.4	8.9	8.0	1.11
	CR60-01-10WB		150	160	0.59	331	34.5	9.3	8.1	1.16
	CR60-01-10DB		150	160	0.59	331	31.8	9.5	8.0	1.18
[36]	BSF4-A100		400	525	2.34	380	26.9	817.6	813.8	1.00
[37]	RAC100-1a		200	268	0.28	640	34.0	26.8	25.2	1.07
	RAC100-2a	100	200	263	1.46	550	34.0	105.4	97.1	1.09
	RAC100-3a		200	244	2.54	550	34.0	142.6	133.7	1.07
[40]	F100-1a		150	200	1.3	572	43.8	41.7	41.9	1.00
	F100-1b		150	200	1.3	572	43.8	41.7	41.9	1.00
	F100-2a		150	200	1.3	572	38.5	44.1	41.3	1.07
	F100-2b		150	200	1.3	572	38.5	42.5	41.3	1.03

Table 2

Statistical descriptors of model factors for Eurocode 2 predictions of beams' flexural strength.

Concrete	Sample size, n	Mean, μ	Standard deviation, σ	CoV (%)	Results outside the 5–95% range
NAC	18	1.064	0.092	8.64	2
RAC50	14	1.079	0.155	14.36	2
RAC100	17	1.091	0.144	13.24	2

samples at the 0.05 significance level. More concretely, this means that EC2 predictions of flexural strength are equally precise and accurate for NAC, RAC50, and RAC100. Flexural strength of RAC beams can be calculated using the existing provisions without any alterations.

3.2. Shear strength of slender beams without stirrups

In this section the predictive capability of Eurocode 2 provisions for shear strength of slender RAC beams without stirrups was

tested. For all the results in database Shear NS the Eurocode 2 predicted values of shear strength were calculated according to the following equation:

$$V_{Rc} = 0.18k(100\rho_l f_c)^{1/3} b_w d \quad (22)$$

where the longitudinal reinforcement ratio is limited to 2% and k is the size effect coefficient:

$$k = 1 + \sqrt{\frac{200}{d}} \leq 2.0 \quad (23)$$

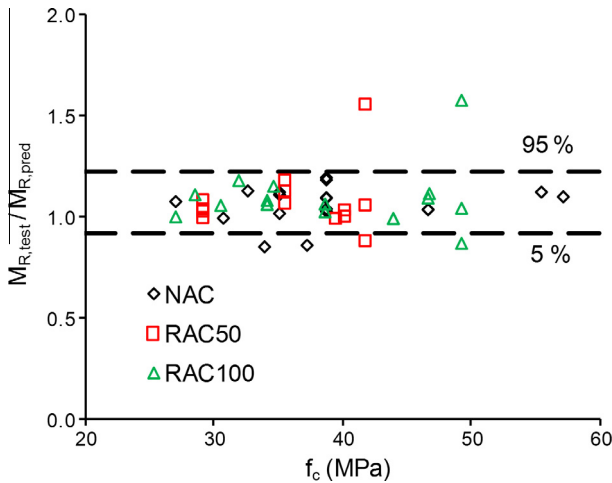


Fig. 5. Model factor for Eurocode 2 predictions of flexural strength compared with concrete compressive strength.

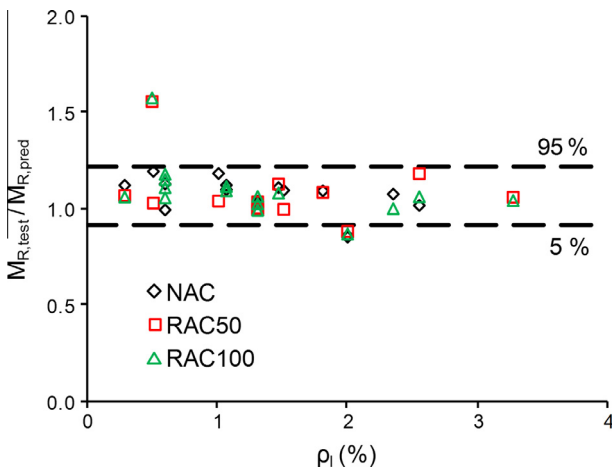


Fig. 6. Model factor for Eurocode 2 predictions of flexural strength compared with longitudinal reinforcement ratio.

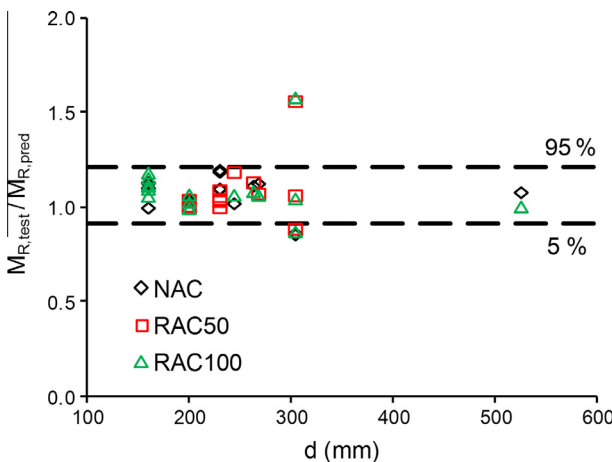


Fig. 7. Model factor for Eurocode 2 predictions of flexural strength compared with beam effective depth.

The shear strength calculated according to Eq. (22) was compared with the maximum allowed value:

$$V_{R,max} = 0.5b_w d f_c \left[0.6 \left(1 - \frac{f_c}{250} \right) \right] \quad (24)$$

and the minimum of the forces was taken as the predicted value $V_{R,pred}$ and the shear model factor, i.e. the test-to-predicted shear strength ratio γ_{sh} was calculated.

The selected values from studies [27–42] that entered the database Shear NS along with relevant data and γ_{sh} are given in Table 3 and the statistical descriptors are given in Table 4. The 507 results from [43] that entered the database are not repeated in Table 3. The mean values for all three samples are very similar and close to 1.

As a first step, Figs. 8–11 present the model factor γ_{sh} in relation to concrete compressive strength, longitudinal reinforcement ratio, cross-section effective depth, and shear span-to-effective depth ratio, respectively. As for flexure, the 5–95% lines were drawn in relation to the γ_{sh} mean value for NAC beams.

Fig. 8 shows practically no correlation between γ_{sh} and concrete compressive strength which means that this parameter is well captured by the current model. All of the NAC values above the 95% line are results from the ACI-DAFStb database. Looking at Figs. 9–11 these outliers can be easily identified. They are beams with a very small effective depth, a very large reinforcement ratio, and a relatively low shear span-to-effective depth ratio. It is possible that for these beams the size effect coefficient k is inadequate and also the limit of 2% for the longitudinal reinforcement ratio imposed by Eq. (22).

What is more important is that all but one RAC results lie within the 5–95% interval meaning that even though a relatively large range of parameters has been studied on RAC beams, they agree with the existing model very well.

As in Section 3.1 this visual analysis was followed up by a statistical one. Again the Kolmogorov-Smirnov normality test was carried out to check whether γ_{sh} for the NAC, RAC50, and RAC100 samples follows the Normal distribution. The test statistics were 0.175 for NAC, 0.163 for RAC50, and 0.229 for RAC100 and the critical values were 0.062, 0.269, and 0.275, respectively. This means that at the 0.05 significance level the null hypothesis should be retained in the case of RAC50 and RAC100 samples but rejected in the case of NAC beams, i.e. the RAC50 and RAC100 samples are normally distributed whereas the NAC sample is not.

The condition for carrying out the t -test is that the samples are normally distributed only when the sample sizes are small (e.g. smaller than 40–50). The Central Limit Theorem states that the average of a large number of independent random variables is approximately normally distributed around the true population mean [47]. With this in mind, although the NAC sample wasn't normally distributed, the t -test was carried out as in the previous section to test whether the means of γ_{sh} for RAC50 and RAC100 samples were equal to that of the NAC sample.

The calculated p -values were 0.377 for the NAC-RAC50 comparison and 0.640 for the NAC-RAC100 comparison. This means that there is no significant difference between the NAC and RAC samples at the 0.05 significance level. As in the case of flexural strength, EC2 predictions for shear strength of beams without stirrups are equally precise and accurate for all concretes. Eq. (22) can be used for RAC beams without stirrups without alterations.

3.3. Shear strength of slender beams with stirrups

The last analyzed case was the EC2 provisions for shear strength of slender RAC beams with stirrups. For all the results in database Shear S the EC2 predicted values of shear strength were calculated according to Eq. (11), section 6.2.3 of EC2. Contrary to MC2010, for predictions according to EC2 the angle θ wasn't calculated but rather it was measured from the photos given in studies [27–42] as the inclination of the critical crack at beam mid-depth. However, its value was restricted to the interval 21.8°–45° as given in the same section.

Table 3

Eurocode 2 predictions of shear strength for 69 selected beams without stirrups.

Study	Specimen	RCA (%)	b_w (mm)	d (mm)	a/d	ρ_l (%)	f_c (MPa)	$V_{E, test}$ (kN)	$V_{R, pred}$ (kN)	γ_{sh}
[28]	HC-1	0	200	303	3.3	2.98	41.9	100.5	86.5	1.16
[30]	V0CC		200	303	3.3	2.98	40.2	88.9	85.3	1.04
[33]	CL-M		200	309	2.6	1.62	38.8	92.8	79.8	1.16
	CG-2.7		200	309	2.6	1.62	34.4	150.0	76.7	1.96
[34]	NANAC-H2.5		200	360	2.5	1.61	24.7	90.7	77.2	1.17
	NANAC-H3.25		200	360	3.25	1.61	24.7	71.1	77.2	0.92
	NANAC-L2.5		200	360	2.5	0.53	24.7	66.2	53.3	1.24
	NANAC-M2.5		200	360	2.5	0.83	24.7	72.0	61.9	1.16
[38]	NAC1b		200	235	4.2	4.09	30.8	106.3	64.2	1.65
[39]	NA-S2		200	300	2.5	1.94	31.8	75.5	77.5	0.97
	NA-M2		200	450	2.5	1.93	31.8	106.9	106.4	1.00
	NA-L2		200	600	2.5	1.94	31.8	125.9	134.5	0.94
	NA-M3		300	450	2.5	2.00	31.8	156.7	161.5	0.97
	NA-L4		400	600	2.5	1.94	31.8	256.4	269.0	0.95
[42]	NAC NS-6 1		300	375	3.2	2.03	37.3	143.2	147.5	0.97
	NAC NS-8 1		300	375	3.2	2.71	37.3	173.5	147.5	1.18
	NAC NS-4 2		300	400	3	1.27	34.2	129.9	129.6	1.00
	NAC NS-6 2		300	375	3.2	2.03	34.2	167.0	143.3	1.17
	NAC NS-8 2		300	375	3.2	2.71	34.2	170.8	143.3	1.19
[40]	S0-1a		150	200	3.8	1.30	32.6	31.1	37.7	0.83
	S0-1b		150	200	3.8	1.30	32.6	36.9	37.7	0.98
	S0-2a		150	200	3.8	1.30	50.3	40.4	43.5	0.93
	S0-2b		150	200	3.8	1.30	50.3	42.3	43.5	0.97
[28]	HR50-1	50	200	303	3.3	2.98	41.3	89.0	86.1	1.03
[30]	V0RC		200	303	3.3	2.98	39.7	90.6	85.0	1.07
[34]	RARAC50-H2.5		200	360	2.5	1.61	24.1	87.9	76.6	1.15
	RARAC50-H3.25		200	360	3.25	1.61	24.1	71.6	76.6	0.93
	RARAC50-M2.5		200	360	2.5	0.83	24.1	67.1	61.4	1.09
[38]	RAC50-1b		200	235	4.2	4.09	33.4	91.8	66.0	1.39
[39]	RH-S2		200	300	2.5	1.94	32.6	60.6	78.1	0.78
	RH-M2		200	450	2.5	1.93	32.6	108.9	107.3	1.01
	RH-L2		200	600	2.5	1.94	32.6	126.1	135.6	0.93
	RH-M3		300	450	2.5	2.00	32.6	154.2	162.9	0.95
	RH-L4		400	600	2.5	1.94	32.6	261.5	271.3	0.96
[42]	RAC50 NS-6 1		300	375	3.2	2.03	32.1	151.3	140.3	1.08
	RAC50 NS-8 1		300	375	3.2	2.71	32.1	171.8	140.3	1.22
	RAC50 NS-6 2		300	375	3.2	2.03	35.5	148.6	145.1	1.02
	RAC50 NS-8 2		300	375	3.2	2.71	35.5	168.7	145.1	1.16
[40]	S50-1a		150	200	3.8	1.30	43.6	44.0	41.5	1.06
	S50-1b		150	200	3.8	1.30	43.6	39.1	41.5	0.94
	S50-2a		150	200	3.8	1.30	40.2	43.7	40.4	1.08
	S50-2b		150	200	3.8	1.30	40.2	41.2	40.4	1.02
[35]	EM-4	63.5	200	305	3.9	2.46	41.6	83.2	86.7	0.96
	EM-L		200	201	2.7	1.99	41.6	89.3	63.0	1.42
	EM-2.7		200	309	2.6	1.62	41.6	103.9	81.7	1.27
	EM-H		200	381	2.7	1.83	41.6	99.5	100.2	0.99
	EM-VH	74.3	200	476	2.7	1.68	41.6	104.6	116.3	0.90
[35]	EV-4		200	305	3.9	2.46	49.1	105.6	91.7	1.15
	EV-L		200	201	2.6	1.99	49.1	122.6	66.6	1.84
	EV-H		200	381	2.7	1.83	49.1	111.7	105.9	1.05
	EV-VH		200	476	2.7	1.68	49.1	119.6	122.9	0.97
[27]	R3.0-N	100	170	270	3	1.10	31.2	55.1	50.0	1.10
[28]	HR100-1		200	303	3.3	2.98	39.8	84.0	85.0	0.99
[34]	RARAC100-H2.5		200	360	2.5	1.61	22.6	84.8	75.0	1.13
	RAC100-M2.5		200	360	2.5	0.83	22.6	70.1	60.1	1.17
[38]	RAC1000-1b		200	235	4.2	4.09	34.5	104.8	66.7	1.57
[39]	RF-S2		200	300	2.5	1.94	34.9	72.9	79.9	0.91
	RF-M2		200	450	2.5	1.93	34.9	96.4	109.8	0.88
	RF-L2		200	600	2.5	1.94	34.9	125.1	138.8	0.90
	RF-M3		300	450	2.5	2.00	34.9	159.8	166.6	0.96
	RF-L4		400	600	2.5	1.94	34.9	256.6	277.5	0.92
[42]	RAC100 NS-6 1		300	375	3.2	2.03	30.0	143.2	137.2	1.04
	RAC100 NS-8 1		300	375	3.2	2.71	30.0	131.4	137.2	0.96
	RAC100 NS-6 2		300	375	3.2	2.03	34.1	124.1	143.2	0.87
	RAC100 NS-8 2		300	375	3.2	2.71	34.1	140.3	143.2	0.98
[40]	S100-1a		150	200	3.8	1.30	41.4	36.4	40.8	0.89
	S100-1b		150	200	3.8	1.30	41.4	38.0	40.8	0.93
	S100-2a		150	200	3.8	1.30	35.7	39.9	38.8	1.03
	S100-2b		150	200	3.8	1.30	35.7	36.1	38.8	0.93

For beams with stirrups EC2 takes the concrete contribution to shear strength into account through this variable inclination of the struts, i.e. the angle θ . However, situations can arise in which the

shear strength without stirrups V_{Rc} is greater than the shear strength with stirrups V_{Rs} . Without going into discussion whether there exist relevant design situations where this can arise, in this

Table 4

Statistical descriptors of model factors for Eurocode 2 predictions of shear strength for beams without stirrups.

Concrete	Sample size, n	Mean, μ	Standard deviation, σ	CoV (%)	Results outside the 5–95% range
NAC	530	1.030	0.279	27.03	75
RAC50	24	1.060	0.151	14.25	0
RAC100	22	1.054	0.233	22.07	2

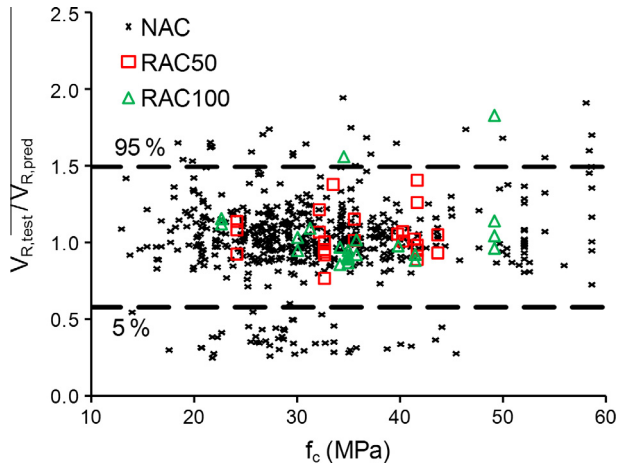


Fig. 8. Model factor for Eurocode 2 predictions of shear strength for beams without stirrups compared with concrete compressive strength.

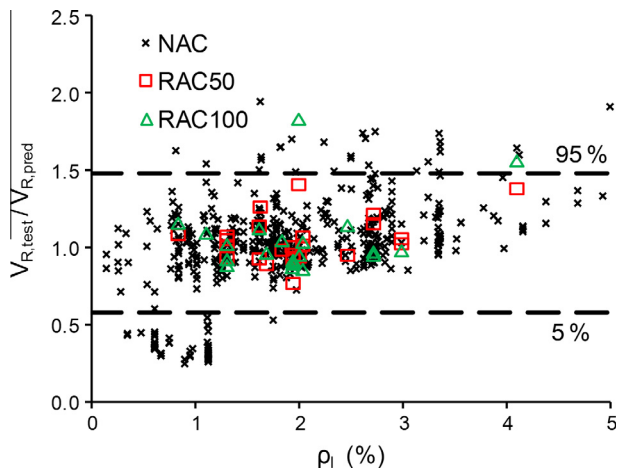


Fig. 9. Model factor for Eurocode 2 predictions of shear strength for beams without stirrups compared with longitudinal reinforcement ratio.

study both values were calculated and compared. The larger of the two was then compared to the maximum allowed value given by:

$$V_{R,max} = \frac{0.6zbf_c}{\cot \theta + \tan \theta} \quad (25)$$

and the minimum of these was taken as the predicted value $V_{R,pred}$ and the shear model factor γ_{sh} was calculated.

As in the previous section, only the 25 values from studies [28–38] that entered the database Shear S along with relevant data and γ_{sh} are given in Table 5 and the statistical descriptors are given in Table 6. The 37 results from [44] that entered the database are not repeated in Table 5.

One very important thing to note is that in 10 out of the 25 results from studies [28–38] the shear strength without stirrups

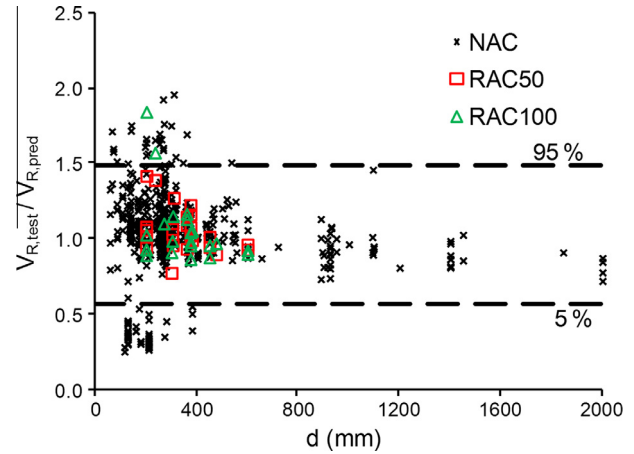


Fig. 10. Model factor for Eurocode 2 predictions of shear strength for beams without stirrups compared with beam's effective depth.

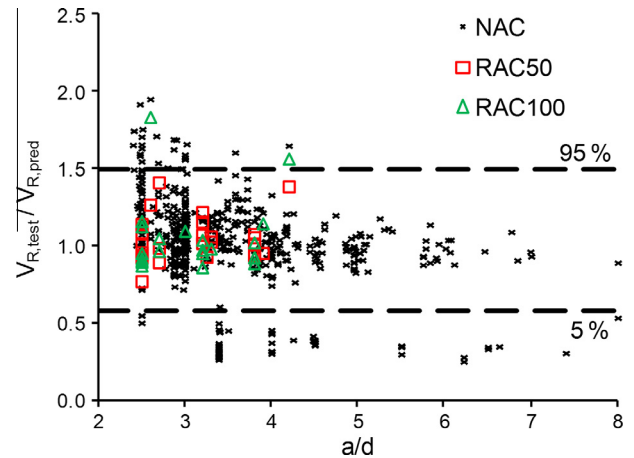


Fig. 11. Model factor for Eurocode 2 predictions of shear strength for beams without stirrups compared with shear span-to-effective depth ratio.

$V_{R,c}$ was larger than the shear strength with stirrups $V_{R,s}$. This is mainly due to the fact that most of those results were beams reinforced with a transverse reinforcement ratio just above the minimum value of $0.08f_c^{0.5}/f_{yw}$. These are obviously beams in which, mechanically, θ would be lower than 21.8° , though this is not allowed by EC2; in other words, the concrete contribution is greater than is allowed for by the code.

This fact is responsible for a large discrepancy between the statistical descriptors of NAC versus RAC50 and RAC100 beams since the majority of NAC results were added from [44] and in those experiments the transverse reinforcement ratios were generally larger. The aforementioned problem is clear in Table 5 where it can be seen that the values of γ_{sh} are very similar for all the concretes, i.e. when only RAC and companion NAC beams are analyzed.

In Figs. 12–15 the model factor γ_{sh} is plotted in relation to concrete compressive strength, unit stirrup stress, cross-section effective depth, and shear span-to-effective depth ratio, respectively. As previously, the 5–95% lines were drawn in relation to the γ_{sh} mean value for NAC beams.

From all the figures a significant upward shift can be seen in the RAC50 and RAC100 results as discussed previously. Perhaps most notably on Fig. 13, the largest values of γ_{sh} are clearly for beams with lower unit stirrup stresses.

Even though the difference between NAC and RAC beams is obvious from Table 6, for the purpose of methodological consistency the

Table 5

Eurocode 2 predictions of shear strength for 25 selected beams with stirrups.

Study	Specimen	RCA (%)	b_w (mm)	d (mm)	a/d	A_{sw} (mm ²)	s (mm)	f_{yw} (MPa)	θ (°)	$V_{E,test}$ (kN)	$V_{E,pred}$ (kN)	γ_{sh}
[28]	HC-2	0	200	303	3.3	57	130	544	19	213.0	161.3	1.32
	HC-3		200	303	3.3	57	170	544	30	177.0	86.5	2.05
	HC-4		200	303	3.3	57	240	544	28	187.5	86.5	2.17
[30]	V24CC		200	303	3.3	57	240	500	22	128.0	84.6	1.51
	V17CC		200	303	3.3	57	170	500	24	150.8	101.9	1.48
	V13CC		200	303	3.3	57	130	500	26	190.3	121.6	1.56
[31]	BNN-lb2		200	250	3.2	57	100	234	30	115.5	68.0	1.70
[38]	NAC3b		200	235	4.2	57	150	300	21	159.9	64.2	2.49
[28]	HR50-2	50	200	303	3.3	57	130	544	28	220.0	121.4	1.81
	HR50-3		200	303	3.3	57	170	544	22	176.0	122.1	1.44
	HR50-4		200	303	3.3	57	240	544	21	164.0	87.4	1.88
[30]	V24RC		200	303	3.3	57	240	500	25	164.3	84.7	1.94
	V17RC		200	303	3.3	57	170	500	35	177.0	86.2	2.05
	V13RC		200	303	3.3	57	130	500	21	233.6	148.3	1.58
[38]	RAC50-3b		200	235	4.2	57	150	300	21	156.9	66.0	2.38
[33]	EM-6S-D	63.5	200	301	2.7	157	200	530	31	341.0	187.7	1.82
[33]	EV-3S-R	74.3	200	301	2.7	101	200	530	27	235.0	141.6	1.66
	EV-6S-D		200	301	2.7	157	200	530	28	327.0	212.1	1.54
[28]	HR100-2	100	200	303	3.3	57	130	544	22	189.5	159.7	1.19
	HR100-3		200	303	3.3	57	170	544	24	163.0	110.8	1.47
	HR100-4		200	303	3.3	101	240	544	29	168.0	112.6	1.49
[31]	ORN-lb2		200	250	3.2	57	100	234	26	118.0	66.2	1.78
	BRN-lb2		200	250	3.2	57	100	234	26	120.5	65.7	1.83
	GRN-lb2		200	250	3.2	57	100	234	26	116.5	67.8	1.72
[38]	RAC100-3b		200	235	4.2	57	150	300	21	163.4	66.7	2.45

Table 6

Statistical descriptors of model factors for Eurocode 2 predictions of shear strength for beams with stirrups.

Concrete	Sample size, n	Mean, μ	Standard deviation, σ	CoV (%)	Results outside the 5–95% range
NAC	45	1.346	0.337	25.03	4
RAC50	8	1.861	0.286	15.34	3
RAC100	9	1.682	0.348	20.71	1

same statistical tests were carried out. The result of the Kolmogorov-Smirnov normality test was 0.178 for NAC, 0.102 for RAC50, and 0.152 for RAC100 and the critical values 0.189, 0.409, and 0.409, respectively. All three samples were normally distributed. The t -test was performed as in the previous sections to test whether the means of γ_{sh} for RAC50 and RAC100 samples are equal to that of the NAC sample.

The calculated p -values were 0.000 for the NAC-RAC50 comparison and 0.009 for the NAC-RAC100 comparison. This means that there is a significant difference between the NAC and RAC samples at the 0.05 significance level and that the null hypothesis of equal means should be rejected. Initially this would suggest that Eq. (11) is not appropriate for RAC beams. However, the discussion in this section rather points to the fact that Eq. (11) is equally inadequate for NAC and RAC beams when they are reinforced with close to minimum transverse reinforcement. Preferably, more studies should be carried out on RAC and companion NAC beams with transverse reinforcement ratios larger than the minimum value.

3.4. Sensitivity analysis

All of the discussion based on the analyses in the previous sections and all of the conclusion drawn from it are dependent on the analyzed databases. They in turn depend primarily on the selection criterion Δ_{cr} , the difference between the test-to-predicted values of shear and flexural strength. In Section 2.2 an argument was proposed why the value $\Delta_{cr} = 0.35$ was chosen. After the analyses it can be seen from the CoVs in Tables 2, 4 and 6 that this choice

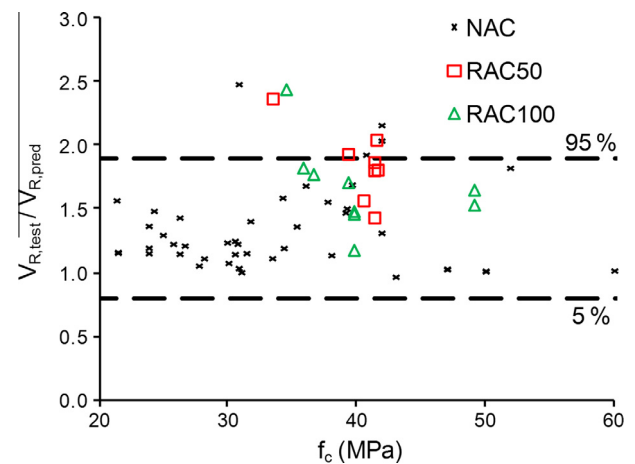


Fig. 12. Model factor for Eurocode 2 predictions of shear strength for beams with stirrups compared with concrete compressive strength.

was adequate. However, it can't be stated with certainty that different results wouldn't have been obtained with different samples.

In order to test the robustness of the conclusions from previous sections, a short sensitivity analysis was carried out. Two additional scenarios are proposed:

- $\Delta_{cr} = 0.25$
- $\Delta_{cr} = 0.45$

With these criteria, formation of new databases Flexure, Shear NS and Shear S was performed. Table 7 presents the results. The number of results in each sample and statistical descriptors are given. In most cases the number of results in each database doesn't vary significantly, i.e. it is not sensitive to the criterion Δ_{cr} . Differences exist for NAC beams in database Shear NS and for RAC100 beams in database Shear S. In the former case the number of results increases or decreases by approximately 80–90 which is around 20% of the initial database. This sample is sensitive to

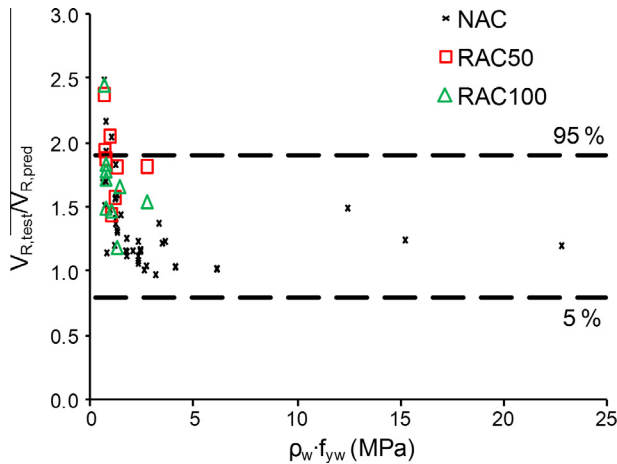


Fig. 13. Model factor for Eurocode 2 predictions of shear strength for beams with stirrups compared with unit stirrup stress.

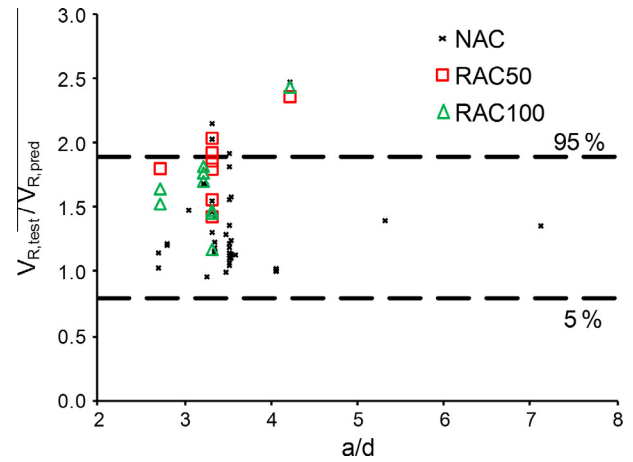


Fig. 15. Model factor for Eurocode 2 predictions of shear strength for beams with stirrups compared with shear span-to-effective depth ratio.

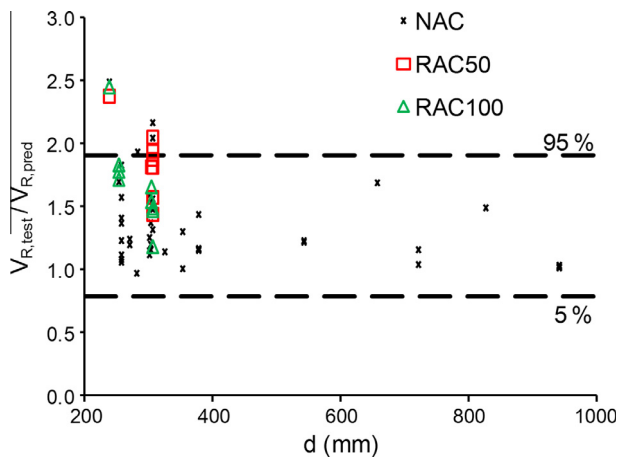


Fig. 14. Model factor for Eurocode 2 predictions of shear strength for beams with stirrups compared with beam's effective depth.

changes in the criterion Δ_{cr} which is to be expected for beams without stirrups. What is important also is that the CoV remains relatively stable around 25%. In the case of RAC100 beams in database Shear S the significant change in mean values is due to the fact that the sample size decreases to only 3 results in the case of $\Delta_{cr} = 0.45$.

The same statistical tests were carried out for the new databases and the only case where there was a change in the results was the NAC-RAC50 comparison in database Shear NS where a p -value of 0.034 was obtained with $\Delta_{cr} = 0.25$. In this case these two samples

are significantly different. Since this is the case with a more relaxed selection criteria (i.e. the test-to-predicted strength ratios can be closer) this result could point to different shear-flexure interaction in RAC beams compared with NAC beams. Further investigation of this topic is not within the scope of this study.

Beside this, the fact that the mean values and CoVs generally don't change significantly for different selection criteria means that the conclusion reached in Section 3 are robust and valid for the current state of knowledge of flexural and shear strength of RAC beams.

4. Conclusions

As with any database, the formation of the one presented in this paper is also subject to bias arising from availability of literature and criteria according to which results are selected. Consequently, all the results from the previous analyses are dependent upon the extensiveness and comprehensiveness of the database. This is why it is important to be transparent about the database creation and analysis process when discussing results and making conclusions.

In this paper, 217 experimental results on RAC and companion NAC beams' flexural and shear strength were gathered from 16 studies. Results were filtered by compressive strength, shear span-to-effective depth ratio, and transverse reinforcement ratio, leaving 194 results. To increase the number of results, already existing databases of NAC beams' shear strengths were added from literature. Within these results, failure types were identified using strict criteria and finally, on each failure type the applicability of Eurocode 2 predictions of flexural and shear strength to RAC beams were tested.

Table 7

Statistical descriptors of model factors for Eurocode 2 predictions of beam's flexural and shear strength for different database selection criteria.

Database	Concrete	$\Delta_{cr} = 0.25$			$\Delta_{cr} = 0.45$		
		n	μ	CoV (%)	n	μ	CoV (%)
Flexure	NAC	23	1.070	7.95	14	1.086	5.63
	RAC50	14	1.079	14.36	14	1.079	14.36
	RAC100	23	1.090	11.50	15	1.088	14.17
Shear NS	NAC	595	1.117	25.50	429	1.177	26.32
	RAC50	27	1.048	14.29	21	1.079	13.25
	RAC100	27	1.034	21.35	18	1.075	23.20
Shear S	NAC	50	1.348	24.67	38	1.375	25.23
	RAC50	8	1.861	15.34	7	1.868	16.49
	RAC100	12	1.683	18.72	3	1.966	24.91

Having this in mind, for the databases created and analyzed in this paper the following conclusions can be drawn:

1. There exist in literature, sufficient experimental results on RAC and companion NAC beams for the creation of a comprehensive database of flexural and shear strengths with 194 results.
2. The failure types (flexural or shear) nominally tested in the studies aren't always achieved in the experiment and criteria must be applied to determine the failure type. This can be done using Model Code 2010 provisions and comparing the difference between the test-to-predicted shear and flexural strength ratios. Using these criteria, out of 194 results, 49 were identified as flexural failure, 69 as shear failure without stirrups, and 25 as shear failure with stirrups while for 51 results the failure type could not be clearly identified.
3. Eurocode 2 predictions of flexural strength are accurate and precise with a mean value of test-to-predicted strength ratio of 1.064 for NAC, 1.079 for RAC50, and 1.091 for RAC100 beams. The CoVs are 8.64%, 14.36%, and 13.24% respectively. Using the statistical *t*-test it was shown that these three samples show no significant difference between them. Flexural strength of RAC beams can be calculated using the existing provisions without any alterations.
4. Eurocode 2 predictions of shear strength for beams without stirrups are accurate but less precise compared to flexural strength. The mean value of test-to-predicted strength ratio is 1.030 for NAC, 1.060 for RAC50, and 1.054 for RAC100 beams while the CoVs are 27.03%, 14.25%, and 22.07% respectively. Using the statistical *t*-test it was shown that these three samples show no significant difference between them. Shear strength of RAC beams without stirrups can be calculated using the existing provisions without any alterations.
5. Eurocode 2 predictions of shear strength for beams with stirrups are both inaccurate and imprecise. The mean value of test-to-predicted strength ratio is 1.346 for NAC, 1.861 for RAC50, and 1.682 for RAC100 beams while the CoVs are 25.03%, 15.34%, and 20.71% respectively. Using the statistical *t*-test it was shown that these three samples are significantly different. This was because most of the experiments on RAC beams were carried out applying close to minimum transverse reinforcement ratios and for this type of beams Eurocode 2 predictions are equally inaccurate and imprecise for both RAC and NAC beams. The difference between NAC and RAC beams arose only when other results on NAC beams from literature were added, with high transverse reinforcement ratios. More experiments on RAC beams with larger than minimum transverse reinforcement ratios should be carried out in order to draw a final conclusion.
6. A sensitivity analysis by selection criteria variation showed that the database of flexural strength is insensitive to criteria variation whereas the databases of shear strengths with and without stirrups are somewhat sensitive. For beams without stirrups this can be explained by a large scatter of the test-to-predicted strength ratio whereas for beams with stirrups the reason is the small number of results on RAC beams.

Acknowledgements

The work reported in this study is a part of the investigation within the Research Project TR36017: Utilization of by-products and recycled waste materials in concrete composites in the scope of sustainable construction development in Serbia: investigation and environmental assessment of possible applications, supported by the Ministry for Education, Science and Technology, Republic of Serbia. This support is gratefully acknowledged.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.conbuildmat.2016.10.058>.

References

- [1] W.H. Langer, J.J. Drew, J.J. Sachs, Aggregate and the Environment, Alexandria, VA, 2004. www.agiweb.org/environment/publications/aggregate.pdf.
- [2] N. Tojo, C. Fischer, Europe as a Recycling Society, Copenhagen, 2011. <http://www.lunduniversity.lu.se/o.o.i.s?id=12683&postid=2303681>.
- [3] P.J. Nixon, Recycled concrete as an aggregate for concrete – a review, *Mater. Struct.* 11 (1978) 371–378.
- [4] C.S. Poon, Z.H. Shui, L. Lam, H. Fok, S.C. Kou, Influence of moisture states of natural and recycled aggregates on the slump and compressive strength of concrete, *Cem. Concr. Res.* 34 (2004) 31–36, [http://dx.doi.org/10.1016/S0008-8846\(03\)00186-8](http://dx.doi.org/10.1016/S0008-8846(03)00186-8).
- [5] M. Etxeberria, E. Vázquez, A. Marí, M. Barra, Influence of amount of recycled coarse aggregates and production process on properties of recycled aggregate concrete, *Cem. Concr. Res.* 37 (2007) 735–742, <http://dx.doi.org/10.1016/j.cemconres.2007.02.002>.
- [6] A. Domingo-Cabo, C. Lázaro, F. López-Gayarre, M.A. Serrano-López, P. Serna, J. O. Castaño-Tabares, Creep and shrinkage of recycled aggregate concrete, *Constr. Build. Mater.* 23 (2009) 2545–2553, <http://dx.doi.org/10.1016/j.conbuildmat.2009.02.018>.
- [7] J. Xiao, B. Lei, C. Zhang, On carbonation behavior of recycled aggregate concrete, *Sci. China Technol. Sci.* 55 (2012) 2609–2616, <http://dx.doi.org/10.1007/s11431-012-4798-5>.
- [8] J. Ying, J. Xiao, L. Shen, M.A. Bradford, Five-phase composite sphere model for chloride diffusivity prediction of recycled aggregate concrete, *Mag. Concr. Res.* 65 (2013) 573–588.
- [9] J. Xiao, J. Li, C. Zhang, Mechanical properties of recycled aggregate concrete under uniaxial loading, *Cem. Concr. Res.* 35 (2005) 1187–1194, <http://dx.doi.org/10.1016/j.cemconres.2004.09.020>.
- [10] L. Evangelista, J. de Brito, Mechanical behaviour of concrete made with fine recycled concrete aggregates, *Cem. Concr. Compos.* 29 (2007) 397–401.
- [11] E.A.B. Koenders, M. Pepe, E. Martinelli, Compressive strength and hydration processes of concrete with recycled aggregates, *Cem. Concr. Res.* 56 (2014) 203–212.
- [12] FIB TG 3.3, Environmental design, in: *FIB Bull.* 28 (2004) 80.
- [13] J. Xiao, T.L. Pham, P.J. Wang, G. Gao, Behaviors of semi-precast beam made of recycled aggregate concrete, *Struct. Des. Tall Spec. Build.* 23 (2014) 692–712, <http://dx.doi.org/10.1002/tal>.
- [14] J. Xiao, C.Q. Wang, J. Li, M. Tawana, Shake-table model tests on recycled aggregate concrete frame structure, *ACI Struct. J.* 109 (2012) 777–786.
- [15] J. Pacheco, J. De Brito, D. Soares, Destructive horizontal load tests of full-scale recycled aggregate concrete structures, *ACI Struct. J.* 112 (2015) 815–826.
- [16] A.M. Knaack, Y.C. Kurama, Sustained service load behavior of concrete beams with recycled concrete aggregates, *ACI Struct. J.* 112 (2015) 565–578, <http://dx.doi.org/10.14359/51687799>.
- [17] K.R.A. Nunes, C.F. Mahler, R. Valle, C. Neves, Evaluation of investments in recycling centres for construction and demolition wastes in Brazilian municipalities, *Waste Manag.* 27 (2007) 1531–1540. <http://linkinghub.elsevier.com/retrieve/pii/S0956053X06002728>.
- [18] S. Marinković, V. Radonjanin, M. Malešev, I. Ignjatović, Comparative environmental assessment of natural and recycled aggregate concrete, *Waste Manag.* 30 (2010) 2255–2264.
- [19] N. Tošić, S. Marinković, T. Dašić, M. Stanić, Multicriteria optimization of natural and recycled aggregate concrete for structural use, *J. Clean. Prod.* 87 (2015) 766–776.
- [20] R.V. Silva, J. de Brito, R.K. Dhir, Tensile strength behaviour of recycled aggregate concrete, *Constr. Build. Mater.* 83 (2015) 108–118, <http://dx.doi.org/10.1016/j.conbuildmat.2015.03.034>.
- [21] R.V. Silva, J. de Brito, R.K. Dhir, Establishing a relationship between the modulus of elasticity and compressive strength of recycled aggregate concrete, *J. Clean. Prod.* (2015), <http://dx.doi.org/10.1016/j.jclepro.2015.10.064>.
- [22] Chinese technical code (CTC), Technical Code on the Application of Recycled Concrete (DG/TJ08-2018-2007), Chinese technical code (CTC), 2007.
- [23] EN 1992-1-1, Eurocode 2: Design of concrete structures – Part 1-1: General rules and rules for buildings, CEN, Brussels, 2004.
- [24] ACI Committee 318, Building Code Requirements for Structural Concrete (ACI 318-11) and Commentary, American Concrete Institute, Farmington Hills, MI, 2011.
- [25] R.V. Silva, J. De Brito, L. Evangelista, R.K. Dhir, Design of reinforced recycled aggregate concrete elements in conformity with Eurocode 2, *Constr. Build. Mater.* 105 (2016) 144–156, <http://dx.doi.org/10.1016/j.conbuildmat.2015.12.080>.
- [26] EN, Eurocode – Basis of Structural Design, CEN, Brussels, 1990, p. 2002.
- [27] B.C. Han, H.D. Yun, S.Y. Chung, Shear capacity of reinforced concrete beams made with recycled-aggregate, in: Fifth CANMET/ACI Int. Conf. Recent Adv. Concr. Technol., 200, 2001, pp. 503–515.
- [28] M. Etxeberria, Experimental study on microstructure and structural behaviour of recycled aggregate concrete, Universitat Politècnica de Catalunya (2004). http://www.tesisenxarxa.net/TDX-0709104-143448/index_cs.html.

- [29] R. Sato, I. Maruyama, T. Sogabe, M. Sogo, Flexural behavior of reinforced recycled concrete beams, *J. Adv. Concr. Technol.* 5 (2007) 43–61, <http://dx.doi.org/10.4334/JKCI.2009.21.4.431>.
- [30] B. Gonzalez-Fonteboa, F. Martinez-Abella, Shear strength of recycled concrete beams, *Constr. Build. Mater.* 21 (2007) 887–893, <http://dx.doi.org/10.1016/j.conbuildmat.2005.12.018>.
- [31] A.B. Ajdukiewicz, A.T. Kliszczewicz, Comparative tests of beams and columns made of recycled aggregate concrete and natural aggregate concrete, *J. Adv. Concr. Technol.* 5 (2007) 259–273, <http://dx.doi.org/10.3151/jact.5.259>.
- [32] G. Fathifazl, A.G. Razaqpur, O.B. Isgor, A. Abbas, B. Fournier, S. Foo, Flexural performance of steel-reinforced recycled concrete beams, *ACI Struct. J.* 106 (2009) 858–867.
- [33] G. Fathifazl, A.G. Razaqpur, O. Burkan Isgor, A. Abbas, B. Fournier, S. Foo, Shear strength of reinforced concrete beams with stirrups, *Mag. Concr. Res.* 62 (2010) 685–699, <http://dx.doi.org/10.1617/s11527-007-9223-3>.
- [34] H.B. Choi, C.K. Yi, H.H. Cho, K.I. Kang, Experimental study on the shear strength of recycled aggregate concrete beams, *Mag. Concr. Res.* 62 (2010) 103–114, <http://dx.doi.org/10.1680/mac.2008.62.2.103>.
- [35] G. Fathifazl, A.G. Razaqpur, O. Burkan Isgor, A. Abbas, B. Fournier, S. Foo, Shear capacity evaluation of steel reinforced recycled concrete (RRC) beams, *Eng. Struct.* 33 (2011) 1025–1033, <http://dx.doi.org/10.1016/j.engstruct.2010.12.025>.
- [36] W. Choi, S.-W. Kim, H.-D. Yun, Flexural performance of reinforced recycled aggregate concrete beams, *Mag. Concr. Res.* 64 (2012) 837–848, <http://dx.doi.org/10.1680/mac.11.00018>.
- [37] I. Ignjatović, S. Marinković, Z. Mišković, A. Savić, Flexural behavior of reinforced recycled aggregate concrete beams under short-term loading, *Mater. Struct.* 469 (2013) 1045–1059.
- [38] I. Ignjatović, Ultimate strength of reinforced recycled concrete beams, University of Belgrade, 2013.
- [39] S.-W. Kim, C.-Y. Jeong, J.-S. Lee, K.-H. Kim, Size effect in shear failure of reinforced concrete beams with recycled aggregate, *J. Asian Archit. Build. Eng.* 12 (2013) 323–330, <http://dx.doi.org/10.3130/jaabe.12.323>.
- [40] A.M. Knaack, Y.C. Kurama, Behavior of reinforced concrete beams with recycled concrete coarse aggregates, *ASCE J. Struct. Eng.* 141 (2014) 1–12, [http://dx.doi.org/10.1061/\(ASCE\)ST.1943-541X.0001118](http://dx.doi.org/10.1061/(ASCE)ST.1943-541X.0001118).
- [41] T.H.-K. Kang, W. Kim, Y.-K. Kwak, S.-G. Hong, Flexural testing of reinforced concrete beams with recycled concrete aggregates (with appendix), *Struct. J.* 111 (2014) 607–616, <http://dx.doi.org/10.14359/51686622>.
- [42] M. Arezoumandi, J. Volz, K. Khayat, Effect of recycled concrete aggregate replacement level on the fracture behavior of concrete, *ACI Mater. J.* 112 (2015) 559–567, <http://dx.doi.org/10.14359/51687766>.
- [43] K.H. Reineck, E. Bentz, B. Fitik, D.A. Kuchma, O. Bayrak, ACI-DaFStb databases for shear tests on slender reinforced concrete beams without stirrups, *ACI Struct. J.* 110 (2013) 867–875, <http://dx.doi.org/10.14359/51686819>.
- [44] K.H. Reineck, E. Bentz, B. Fitik, D.A. Kuchma, O. Bayrak, ACI-DaFStb databases for shear tests on slender reinforced concrete beams with stirrups, *ACI Struct. J.* 111 (2014) 1147–1156, <http://dx.doi.org/10.14359/51686819>.
- [45] CEB-FIP Model Code 2010 Volume 2, International Federation for Structural Concrete (fib), 2010.
- [46] A. Marí, J.M. Bairán, A. Cladera, E. Oller, Shear design and assessment of reinforced and prestressed concrete beams based on a mechanical model, *ASCE J. Struct. Eng.* 142 (10) (2016) 1–17, 04016064, [http://dx.doi.org/10.1061/\(ASCE\)ST.1943-541X](http://dx.doi.org/10.1061/(ASCE)ST.1943-541X).
- [47] T. Lumley, P. Diehr, S. Emerson, L. Chen, The importance of the normality assumption in large public health data sets, *Annu. Rev. Public Heal.* 23 (2002) 151–169, <http://dx.doi.org/10.1146/annurev.publhealth.23.100901.140546>.